

The examples we have given above are possibly the simplest attacks that can be made to a server. Most exploits publicly available on websites such as milw0rm are actually available as Perl / Python / PHP / C++ etc., and might sometimes contain intentionally introduced but obvious flaws, in order to discourage usage by people who just wish to download and run such exploits for fun without any knowledge of what they are doing.

Even for these simple attacks, you needed to know about some core OS features such as alternate data streams. You should be able to understand now that it is not a simple task to create such exploits – and these are some of the simplest ones.

In any case, there is a big world out there with some people just looking to have some fun at your expense. With this knowledge of the vulnerabilities your infrastructure faces, you can begin to be prepared. There is no perfect defence, however one must do all they can. Always keep the internet-facing computers updated with the latest patches, to ensure that there is minimal security risk.

While it is impossible to impart a thorough knowledge of penetration testing in such a short book, we hope that what you have now is a good foundation to start understanding some hacking concepts.

3.2 Web application hacking

3.2.1 Introduction

In the previous section, we covered how we could hack network services and possibly gain access to the computers, or its data. Now we will talk about the vulnerabilities of web applications, and how they can be exploited.

The internet is accessible to everyone, and connects everyone. However, it also makes governing it nearly impossible. You never know where your next attack is coming from, and often you might not even know about the vulnerabilities of a system until it is attacked.

The applications you run on your server are accessible to everyone, and can be attacked by everyone. In such a distributed medium, it is simple to find vulnerabilities and easier still to make them public. Even as a vulnerability gets patched, the updates slowly trickle to all web servers using the application. Odds are you will find many are still using older version with the vulnerabilities unpatched. While this might not be of much use to people who already have a particular target in mind, there are many who simply looking for an easy hack without concern of who it is affecting.

That said it is important that the applications you run do not expose such vulnerabilities, because those who attack you are not necessarily your enemies or those who stand to gain something from the attack, but include those who are doing so just for fun.

An increasing number of websites are becoming dynamic. Few websites